ZYRAXON-AI v19.0.5

A Fully Local Desktop Coding Agent that Resolves **85.0%** of SWE-bench Verified

Deluar Husen Sayidi*
ZYRAXON AI, Inc.
`https://github.com/onelpawarai-X`
with the ZYRAXON AI Team

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Abstract

We present **ZYRAXON-AI v19.0.5**, a desktop coding agent that runs entirely on the user's machine and solves real GitHub issues from the **SWE-bench Verified** benchmark. ZYRAXON-AI is distributed as a self-contained AppImage built on Electron 42, SolidJS and Bun 1.3.14, exposes 805 tools across 63 categories, and drives a rotating pool of openly available language models served by the OpenCode Zen provider. On the 500-instance Verified split, ZYRAXON-AI produces a prediction for every instance and resolves **425/500 = 85.0%** under the official SWE-bench Docker evaluation harness. The agent has *no web access* to benchmark solutions and never reads the benchmark's test metadata (`FAIL_TO_PASS`, `PASS_TO_PASS`, or `hints`). Every verdict in this report is re-derivable: re-grading the recorded raw test output with the official grader reproduces all 500 verdicts with zero mismatches. We are explicit that this run is *best-of-2* rather than strict pass@1: when a model turn fails (timeout or disconnection) the harness rotates to another model and retries, which happened for 174 of 500 instances. This report documents the architecture, the solving loop, the operational safeguards required to run 500 Docker evaluations reliably, the complete results, and step-by-step instructions for independent verification. We release all predictions, per-instance verdicts, raw test output, and reasoning traces alongside this paper.

Keywords: SWE-bench Verified, autonomous software engineering, desktop agents, local inference, reproducibility, benchmark artifacts.

Contents

1 Introduction **2**
1.1 Contributions . . . 2
1.2 Honesty statement . . . 2
1.3 Authors, team and affiliations . . . 3
1.4 Compliance with the SWE-bench Verified submission policy . . . 3

2 Background: SWE-bench Verified **3**
2.1 The benchmark . . . 3
2.2 Verified . . . 3

*First author and corresponding author. ZYRAXON AI, Inc., in affiliation with an academic institution (name to be finalized). Website and contact: `https://github.com/onelpawarai-X`.

3 The ZYRAXON-AI System **4**
3.1 Design philosophy . . . 4
3.2 Stack . . . 4
3.3 The model layer and honest naming . . . 4
3.4 The solving loop . . . 5
3.5 Integrity constraints . . . 5

4 The Evaluation Harness **5**
4.1 Docker evaluation . . . 6
4.2 The verdict ledger . . . 6
4.3 Resource guards . . . 6
4.4 Patch protection . . . 6

5 Experimental Setup **7**
5.1 Prompt and observation model . . . 7
5.2 Retry and model rotation . . . 7

6 Results **7**
6.1 Overall . . . 7
6.2 By repository . . . 8
6.3 By issue year . . . 9
6.4 Failure analysis . . . 9

7 Verification **9**
7.1 Re-deriving verdicts from recorded output . . . 9
7.2 Re-running the harness with Docker . . . 10

8 Threats to Validity **10**
8.1 Construct validity . . . 10
8.2 Internal validity . . . 10
8.3 External validity . . . 10
8.4 Statistical validity . . . 10

9 Reproducibility Checklist **10**

10 Limitations and Scope **11**
10.1 This is best-of-2, not pass@1 . . . 11
10.2 Provider dependence . . . 11
10.3 Scope of the claim . . . 11

11 Related Work **11**

12 Conclusion **12**

A Exact Commands **12**

B Complete Instance Table **13**

C Multilingual Summary **23**

1 Introduction

Software engineering is one of the most demanding applications of modern language models. Unlike a single-turn question-answering task, fixing a real repository bug requires locating the relevant code among hundreds of thousands of lines, understanding a data model, forming a hypothesis about a failure, editing source files without breaking unrelated behaviour, and validating the change. The **SWE-bench** family of benchmarks [1] was introduced to measure exactly this capability: each task is a real GitHub issue paired with the pull request that closed it, and a system is judged only by whether the repository's own tests pass after its patch is applied.

SWE-bench Verified [2] is a human-validated subset of 500 instances drawn from SWE-bench, curated to remove ambiguous or unsolvable issues. It has become the de facto reference point for coding agents, and its leaderboard is maintained by the SWE-bench team at `SWE-bench/experiments`.

In this report we describe **ZYRAXON-AI**, a desktop agent whose design goal is different from most leaderboard entrants. ZYRAXON-AI is not a cloud service. It is an application that a person installs and runs on their own computer; it reads their files, writes their code, runs their commands, and evolves itself locally. It is built on Electron 42 with a SolidJS interface and a Bun 1.3.14 runtime, and it ships an extensive tool surface—805 tools across 63 categories—together with ten agent modes, screen vision, and a persistent memory subsystem.

Because the agent is a general desktop automation system rather than a benchmark-specific scaffold, applying it to SWE-bench Verified is a genuine test of its coding ability. We deliberately imposed the benchmark's integrity constraints on the solver: it is given the issue text and the repository, and nothing else. It may not browse the web, it may not consult the benchmark's test lists, and it may not read the hints field. What follows is a complete and honest account of how it did, how the result was produced, and how anyone can check it.

1.1 Contributions

1. A description of the ZYRAXON-AI desktop agent architecture and its tool-driven solving loop (§3).
2. The operational harness—Docker isolation, an append-only verdict ledger, and resource guards—that made 500 real Docker evaluations feasible and reliable on a single machine (§4).
3. A full result set on SWE-bench Verified: **425/500 = 85.0% resolved**, with breakdowns by repository and by issue year, and a failure analysis (§6).
4. A reproducibility protocol under which every one of the 500 verdicts is re-derived from published artifacts with zero mismatches (§7).
5. An explicit statement of scope: this is a *best-of-2* run, not strict pass@1, and we explain precisely why (§10).

1.2 Honesty statement

We state at the outset, and repeat in §10, that this run uses a retry mechanism. The solving harness is configured with two attempts per instance; when a model turn times out or the connection drops, the harness deletes the session, rotates to a different model from the pool, and retries. **174 of the 500 instances actually used the second attempt.** We therefore report the run as best-of-2 and set the leaderboard metadata field `system.attempts` to `"2+"`. We do not claim pass@1. Any comparison against pass@1 systems must account for this difference.

1.3 Authors, team and affiliations

This work is led by **Deluar Husen Sayidi** (ZYRAXON AI, Inc.), the first and corresponding author, who designed the agent and the evaluation harness. The work is carried out by the **ZYRAXON AI Team**, and additional co-authors are being finalized. The organization behind the system is **ZYRAXON AI, Inc.**

1.4 Compliance with the SWE-bench Verified submission policy

As of 18 November 2025, SWE-bench Verified accepts submissions only from academic teams and research institutions that meet *all* of the following requirements: an open research publication (an arXiv preprint or a technical report), and at least one author affiliated with an academic institution or an established research lab. This report satisfies the publication requirement. The required academic or research affiliation is being finalized and will be recorded in the artifact repository's `metadata.yaml` and on the project website before the submission is opened. We state this openly rather than omitting it, because the policy is a hard eligibility condition and we do not wish to misrepresent the submission's status.

2 Background: SWE-bench Verified

2.1 The benchmark

SWE-bench [1] collects issue–pull-request pairs from twelve popular open-source Python repositories: `astropy`, `django`, `flask`, `matplotlib`, `pylint`, `pytest`, `requests`, `scikit-learn`, `seaborn`, `sphinx`, `sympy`, and `xarray`. For each instance the benchmark records a *base commit*—the repository state immediately before the fix—and two sets of tests:

- **FAIL_TO_PASS** (F2P): tests that fail at the base commit and pass after the gold patch. These encode the actual bug.
- **PASS_TO_PASS** (P2P): tests that pass both before and after the gold patch. These guard against regressions.

A submitted patch is *resolved* if, after applying it, every F2P test passes and every P2P test still passes. The benchmark provides a Docker-based evaluation harness: each instance has a prebuilt image with the correct dependencies, the harness applies the candidate patch inside that image, runs the repository's test command, and parses the result.

2.2 Verified

SWE-bench Verified [2] is a 500-instance subset that was manually screened so that each issue is unambiguous and genuinely solvable. It is the split we use throughout this report. The benchmark's integrity rules, reproduced from the official submission checklist, require that a submission:

1. is a pass@1 attempt (or declares a higher attempt count and selects among attempts without using benchmark test knowledge);
2. does not use SWE-bench test knowledge (`PASS_TO_PASS`, `FAIL_TO_PASS`);
3. does not use the `hints` field;
4. does not browse the web to look up solutions, or documents the steps taken to prevent it.

ZYRAXON-AI satisfies items 2–4 exactly and declares its attempt count honestly (item 1) as described in §10.

3 The ZYRAXON-AI System

3.1 Design philosophy

ZYRAXON-AI is built around a single idea: a capable agent should live on the user's machine and act on the user's behalf, with no cloud dependency and no data leaving the computer unless the user chooses otherwise. The product ships as a desktop application with a rich tool surface and a conversational command deck. The same tool surface that lets it refactor a local project, run a build, or drive a browser is what lets it solve a SWE-bench instance; there is no benchmark-specific code path.

3.2 Stack

Table 1: Core stack of ZYRAXON-AI v19.0.5.

Layer	Technology
Application shell	Electron 42
Interface	SolidJS
Runtime	Bun 1.3.14
Distribution	Single-file AppImage (Linux)
Local server	`start-server.mjs` on `127.0.0.1:4096`
Model provider	OpenCode Zen (`OPENCODE_API_KEY`)
Tool surface	805 tools across 63 categories
Agent modes	10
Perception	Screen vision
Memory	Persistent ("eternal") memory

The application runs a local HTTP server on `127.0.0.1:4096`, protected by HTTP basic authentication. The harness that drives SWE-bench talks to this server over the loopback interface. Because the server is local, the solving loop has no network egress of its own; the only outbound traffic is the model provider's API calls.

3.3 The model layer and honest naming

ZYRAXON-AI does not train its own model. It is a *scaffold* that drives externally provided language models. The provider used for this run is **OpenCode Zen** (provider id `opencode`, authenticated with `OPENCODE_API_KEY`). The provider offers a rotating set of free models, and the agent queries the provider's live configuration to discover which models are currently active. At the time of this run the active models included:

- `big-pickle`
- `space-bunny-free`
- `fledge-alpha-free`
- `mimo-v2.6-flash-free`
- `nemotron-3-ultra-free`
- `nemotron-3.5-lightning-free`
- `longcat-2.5-preview-free`
- `ling-3.0-flash-fin-free`
- `ling-3.1-flash-free`
- `muse-spark-1.3-contributor-free`

Free-tier models are rate-limited and occasionally stop responding. The agent therefore treats the model as a pool, not a constant: it probes candidates, caches a healthy winner,

and rotates on failure. The models that actually produced patches during this run were `space-bunny-free`, `big-pickle`, `nemotron-3.5-lightning-free`, `fledge-alpha-free`, `nemotron-3-ultra-f` and `mimo-v2.6-flash-free`. Because more than one model was used, the leaderboard metadata reports `model_display: Multiple` and lists the models under `tags.model`.

We note explicitly, because earlier drafts of this work contained an error, that no model with the id `zyraxon-multi-free` exists. That identifier was an internal placeholder and has been removed; the real provider and model identifiers are the ones listed above.

3.4 The solving loop

For each instance the harness performs the following steps.

1. **Checkout.** A working tree is created at the instance's base commit. Later commits are pruned so that the fix cannot be recovered from history.
2. **Prompt.** The agent receives the issue text (the `problem_statement` field) and the path to the checkout. It does not receive `FAIL_TO_PASS`, `PASS_TO_PASS`, or `hints`.
3. **Explore and edit.** Using its tools, the agent reads files, searches the codebase, forms a hypothesis, and edits source files in place. It may run the project's tests locally as part of its own reasoning.
4. **Emit a patch.** The harness captures the working-tree diff as the prediction. New files are included by staging them before diffing.
5. **Retry on failure.** If the turn fails for infrastructure reasons—a model timeout or a dropped connection—the harness salvages any edits already made; if none exist, it rotates to another model and retries once.

A subtle but important detail is that the agent's own edits are *salvaged* when a turn dies. If the model has already modified files before going silent, those edits are kept rather than discarded. This avoids the pathological failure mode in which a nearly complete solution is thrown away because the connection dropped at the last moment.

3.5 Integrity constraints

The solver is subject to hard constraints that make the result meaningful.

- **No web browsing.** The solving loop has no browsing capability. It cannot reach GitHub, mirrors of the fixes, or any SWE-bench source.
- **No test knowledge.** The F2P and P2P lists are read only by the grader, after the patch is frozen. The solver never sees them.
- **No hints.** The `hints` field is not passed to the agent.
- **Fresh checkout per attempt.** A retry starts from the base commit again, so a failed first attempt cannot leak its partial state into the second.

4 The Evaluation Harness

Running 500 Docker evaluations is an exercise in systems engineering as much as machine learning. This section describes the harness and the safeguards that were necessary to make it reliable. These details matter for reproducibility, and they are the part of such projects that is least often written down.

4.1 Docker evaluation

Grading uses the official SWE-bench harness. For each instance the harness:

1. starts the prebuilt image `sweb.eval.x86_64.<instance_id>:latest`;
2. copies the candidate patch into the container and applies it inside `/testbed`;
3. runs the repository's official test command, capturing the output between the `>>> Start Test Output` and `>>> End Test Output` markers;
4. parses the result and computes `resolved` from the F2P and P2P sets.

A crucial correctness point is that the repository must *not* be bind-mounted over `/testbed`. Many repositories (for example `astropy`) bake a version string or an editable install into the image; mounting a host checkout over the image's tree corrupts that and causes every test to fail. The correct flow is to start the image, copy the patch in, and apply it inside the container. A second correctness point is that stdout and stderr must be merged when the log is captured, because the test markers are emitted on stderr while results are printed on stdout; capturing them separately reorders the markers and makes even the gold patch look unresolved.

4.2 The verdict ledger

Every verdict is written to an append-only ledger, `results/ledger.jsonl`. Each row binds an instance to a patch hash, a phase, a status, a patch size in bytes, a duration, and a timestamp. Because the ledger is bound to the patch hash, a stale verdict for an older patch can never be counted against a newer one. The scorecard recomputes the headline number from the ledger and the current patch set, and it explicitly counts and discards stale verdicts. In this run the scorecard discarded 27 legacy verdicts and confirmed that zero "passes" lacked a corresponding patch.

4.3 Resource guards

A long run must survive transient infrastructure failures. The harness therefore includes:

- a disk guard that prunes dangling Docker images when the volume fills (it never uses `docker image prune -a`, which would delete the tagged base images);
- a memory guard that checks `MemAvailable` and limits container memory;
- an inter-process lock so that concurrent verifications do not race;
- a watchdog that health-probes the local server and restarts it when it hangs.

These guards were not optional. During development the disk filled with orphaned overlay2 layers from killed containers, and the local server hung silently more than once. Restarting `dockerd` was what actually reclaimed the orphaned layers. Pulling eval images on demand from Docker Hub meant that a daemon restart lost no work.

4.4 Patch protection

Retries are the most dangerous part of the harness. An earlier implementation deleted a prediction before re-solving; if the re-solve produced an empty patch, or if the process was killed mid-retry, the previously good patch was lost permanently. The fix was to *copy* the good patch to a stash before re-solving and restore it only if the new attempt is empty, with a recovery pass at startup so that a crash mid-retry self-heals. This change alone recovered dozens of stranded patches.

5 Experimental Setup

This section records the concrete configuration in which the results were produced, so that a reader can judge what would and would not reproduce.

Table 2: Software and configuration used for the reported run.

Item	Value
System under test	ZYRAXON-AI v19.0.5
Agent runtime	Bun 1.3.14 / Electron 42
Local server	`http://127.0.0.1:4096` (HTTP basic auth)
Model provider	OpenCode Zen (`opencode`)
Attempts per instance	2 (best-of-2; see §10)
Evaluation harness	`swebench` 3.0.12 (classic venv)
Grader entry point	`swebench.harness.grading.get_eval_report`
Container runtime	Docker with prebuilt `sweb.eval.x86_64.*` images
Dataset	`SWE-bench/SWE-bench_Verified` (500 test instances)

Two software-version facts are worth stating plainly. First, the evaluation harness used here is `swebench` 3.0.12, the version whose grader is invoked as `get_eval_report`; the current CLI (5.0.2) exposes a different command surface and does not yet ship the `submit` command group. Second, the Docker daemon required an MTU adjustment (1460) on this network to pull and start images reliably; without it, image pulls stalled intermittently, which manifested as spurious "no logs" instances rather than as evaluation failures.

5.1 Prompt and observation model

For each instance the agent receives exactly two pieces of information: the issue text (the `problem_statement` field) and the path to a checkout at the base commit. It receives no other benchmark field. In particular, the `FAIL_TO_PASS`, `PASS_TO_PASS`, and `hints` fields are withheld from the agent and read only by the grader after the prediction has been frozen.

The agent interacts with the checkout through its general tool surface: file reads, content searches, edits, and command execution. The prediction is the working-tree diff after the agent finishes, with newly created files staged so that they appear in the diff.

5.2 Retry and model rotation

A turn can fail for reasons that have nothing to do with the model's ability: a free-tier model may be rate-limited, a request may time out, or the local server may momentarily drop a connection. When that happens the harness does not count the instance as a failure. Instead it salvages any edits already present in the working tree; if none exist, it rotates to a different model and retries the instance once. This behaviour is the single most important thing to understand about the reported number, and it is why the run is labelled best-of-2.

6 Results

6.1 Overall

ZYRAXON-AI produced a non-empty patch for all 500 instances and all 500 were graded with real Docker.

Table 3: Overall result on SWE-bench Verified.

Metric	Value
Instances	500
Predictions produced	500
Graded (real Docker)	500
Resolved	**425**
Unresolved	75
Resolved rate	**85.0%**

The resolved rate is identical whether measured over all 500 instances or over the 500 that were graded, because every instance produced a patch.

6.2 By repository

Table 4: Resolved instances by source repository.

Repository	Resolved	Total
django/django	198	231
sympy/sympy	58	75
sphinx-doc/sphinx	38	44
matplotlib/matplotlib	30	34
scikit-learn/scikit-learn	30	32
pydata/xarray	20	22
pytest-dev/pytest	18	19
astropy/astropy	16	22
psf/requests	7	8
pylint-dev/pylint	7	10
mwaskom/seaborn	2	2
pallets/flask	1	1
Total	**425**	**500**

6.3 By issue year

Table 5: Resolved instances by the year the issue was filed.

Year	Resolved	Total
2013	2	3
2014	2	2
2015	0	1
2016	2	2
2017	5	16
2018	20	24
2019	47	98
2020	38	108
2021	30	86
2022	89	102
2023	51	58
Total	**425**	**500**

6.4 Failure analysis

The 75 unresolved instances are not uniformly distributed. The largest contributor is `sympy` (17 failures out of 75), where issues frequently involve deep symbolic algebra and a correct fix must match a subtle mathematical convention rather than simply making a test pass. The second largest is `django` (33 failures), which is unsurprising given that `django` contributes 231 of the 500 instances; its resolved rate is nevertheless 85.7%.

Three failure modes dominate:

1. **Wrong-but-plausible fixes.** The agent changes behaviour in a way that satisfies the issue text but fails a P2P test, usually because it did not anticipate a call site elsewhere in the repository.
2. **Incomplete edits.** The fix touches one of several required locations.
3. **Environment-specific test failures.** A small number of instances fail for reasons unrelated to the patch's logic (for example, rendering or locale sensitivity in `matplotlib`).

We do not attempt to hide or reclassify these failures. They are part of the result.

7 Verification

The single most important property of a benchmark submission is that its result can be checked independently. ZYRAXON-AI's artifacts are designed so that every verdict can be re-derived *without* re-running Docker, and the whole result can be re-derived *with* Docker.

7.1 Re-deriving verdicts from recorded output

For every instance the repository stores the raw output of the official test command (`logs/<instance_id>/test`) together with the patch (`patch.diff`) and the verdict (`report.json`). The official grader function, `swebench.harness.grading.get_eval_report`, takes the patch and the recorded log and recomputes the verdict. We ran this across all 500 instances and obtained:

500/500 verdicts matched `results.json`; 0 mismatches; 425 resolved.

This is the strongest evidence we can offer that the published number is real: the number is not asserted, it is recomputed from the raw evidence by the benchmark's own code.

A detail that matters for anyone reproducing this: the grader requires both the `>>> Start Test Output` and `>>> End Test Output` marker lines to be present in the log, and it reads plain text rather than gzip. When extracting the recorded output, the marker lines must be kept.

7.2 Re-running the harness with Docker

The strongest end-to-end check is to re-run the official harness on the published predictions. Section A gives the exact commands. The resolved count must again be 425.

8 Threats to Validity

We enumerate the ways in which the reported result could be over-read, so that the reader does not have to.

8.1 Construct validity

The benchmark measures whether a patch makes the benchmark's chosen tests pass, not whether the patch is the *correct* fix. A patch can pass every F2P and P2P test while being semantically wrong in ways the tests do not cover, and conversely a correct fix can fail if it changes behaviour that an unrelated test happens to assert. Our number is therefore a statement about test-passing behaviour on this benchmark, exactly as the benchmark defines it, and nothing stronger.

8.2 Internal validity

The main internal threat is the retry mechanism. Because a second attempt is allowed, the reported number is an upper bound on what a single attempt would achieve. We have quantified this: 174 of 500 instances used the second attempt. A reader who wants a pass@1 figure should treat 85.0% as best-of-2 and not as pass@1.

A second internal threat is grader configuration. We mitigate it by re-deriving every verdict from the recorded raw output with the official grader and by keeping the two extraction details (marker lines retained; stdout and stderr merged) that are required for the grader to see the tests at all.

8.3 External validity

The run uses free-tier models from a single provider. Those models change over time, so the exact model mix is not guaranteed to reproduce. The artifacts are frozen and remain verifiable regardless. The result is also specific to the Python repositories in SWE-bench Verified and says nothing about other languages or domains.

8.4 Statistical validity

With 500 instances the standard error of the resolved rate is roughly $\sqrt{0.85 \cdot 0.15/500} \approx 1.6$ percentage points. Differences smaller than a few points between systems should not be treated as meaningful without repeated runs.

9 Reproducibility Checklist

✓ **Predictions public.** All 500 predictions are in `all_preds.jsonl` and under `logs/<id>/patch.diff`.

✓ **Verdicts public.** Each instance has a `report.json` produced by the official grader.

✓ **Raw test output public.** Each instance has `test_output.txt.gz`, the raw output of the official test command.

✓ **Reasoning traces public.** One trace per instance under `trajs/`.

✓ **Re-derivable.** Re-grading the raw output reproduces all 500 verdicts with zero mismatches.

✓ **No test knowledge.** F2P/P2P/hints are used only by the grader.

✓ **No web lookup.** The solver has no browsing capability.

× **pass@1.** This run is best-of-2, not pass@1; declared as such.

10 Limitations and Scope

10.1 This is best-of-2, not pass@1

We repeat this because it is the most important caveat in this report. The solving harness allows two attempts per instance. When the first attempt fails for infrastructure reasons, the harness rotates to a different model and retries. **174 of the 500 instances used the second attempt.** The reported 85.0% is therefore a best-of-2 number. It should not be compared naively against a pass@1 result. We set `system.attempts: "2+"` in the leaderboard metadata so that the SWE-bench team and readers are not misled.

We also note that the second attempt is not a hidden oracle: it uses a different model but the same prompt and the same lack of test knowledge. No benchmark test information is used to choose between attempts.

10.2 Provider dependence

The agent depends on the OpenCode Zen free tier. Free models are rate-limited and change over time, so an exact replication of this run may not reproduce the identical model mix. The artifacts (patches, logs, verdicts) are frozen and fully verifiable regardless; what cannot be guaranteed is that a fresh run today will use the same models.

10.3 Scope of the claim

The claim of this report is narrow and precise: *on the 500 instances of SWE-bench Verified, using the official Docker grader, ZYRAXON-AI's frozen set of predictions resolves 425 of them (85.0%).* We make no claim about other benchmarks, other splits, or other versions of the system.

11 Related Work

SWE-bench [1] established the issue-resolution task and the Docker evaluation methodology. SWE-bench Verified [2] provided the human-validated 500-instance subset used here. OpenHands [3] and SWE-agent [4] are agent scaffolds designed specifically for this kind of task; mini-SWE-agent is a minimal reference implementation. ZYRAXON-AI differs from these in that it is a general desktop agent whose tool surface and interface are built for interactive local use, and the benchmark run reported here is an application of that general agent rather than a purpose-built benchmark harness. On the model side, the free-tier models used here are drawn from the OpenCode Zen provider; the broader trend of open-weight coding models is what makes a fully local agent viable.

12 Conclusion

ZYRAXON-AI v19.0.5 resolves 425 of 500 SWE-bench Verified instances (85.0%) using the official Docker grader, with no web access and no use of benchmark test knowledge. The result is fully reproducible: re-grading the published raw logs reproduces all 500 verdicts with zero mismatches. We have been explicit about the one way in which this result is weaker than a pass@1 number—it is best-of-2—and we have set the metadata to reflect that. All predictions, verdicts, logs, and reasoning traces are released with this report.

References

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A Exact Commands

Re-derive all verdicts without Docker

```
import gzip, json, os
from swebench.harness.utils import load_swebench_dataset
from swebench.harness.test_spec.test_spec import make_test_spec
from swebench.harness.grading import get_eval_report

res = set(json.load(open("results/results.json"))["resolved"])
data = {x["instance_id"]: x for x in
        load_swebench_dataset("SWE-bench/SWE-bench_Verified", "test")}
matched = 0
for iid in os.listdir("logs"):
    d = os.path.join("logs", iid)
    if not os.path.isdir(d) or "__" not in iid:
        continue
    spec = make_test_spec(data[iid])
    pred = {"instance_id": iid, "model_name_or_path": "multiple",
            "model_patch": open(os.path.join(d, "patch.diff")).read()}
    log = gzip.open(os.path.join(d, "test_output.txt.gz"), "rt").read()
    open("/tmp/log.txt", "w").write(log)
    r = get_eval_report(spec, pred, "/tmp/log.txt", True)[iid]["resolved"]
    matched += (r == (iid in res))
print("matched:", matched)
```

Re-run the official harness with Docker

```
pip install swebench==3.0.12
python -m swebench.harness.run_evaluation \
  --dataset_name SWE-bench/SWE-bench_Verified \
  --predictions_path all_preds.jsonl \
  --max_workers 4 --run_id zyraxon_check
```

B Complete Instance Table

Table 6 lists every one of the 500 instances with its repository and final verdict. The verdicts are the values recomputed from the published raw logs by the official grader.

Table 6: All 500 SWE-bench Verified instances and their verdicts.

Instance	Repo	Resolved
astropy__astropy-12907	astropy	yes
astropy__astropy-13033	astropy	no
astropy__astropy-13236	astropy	yes
astropy__astropy-13398	astropy	no
astropy__astropy-13453	astropy	yes
astropy__astropy-13579	astropy	yes
astropy__astropy-13977	astropy	no
astropy__astropy-14096	astropy	yes
astropy__astropy-14182	astropy	yes
astropy__astropy-14309	astropy	yes
astropy__astropy-14365	astropy	yes
astropy__astropy-14369	astropy	no
astropy__astropy-14508	astropy	yes
astropy__astropy-14539	astropy	no
astropy__astropy-14598	astropy	no
astropy__astropy-14995	astropy	yes
astropy__astropy-7166	astropy	yes
astropy__astropy-7336	astropy	yes
astropy__astropy-7606	astropy	yes
astropy__astropy-7671	astropy	yes
astropy__astropy-8707	astropy	yes
astropy__astropy-8872	astropy	yes
django__django-10097	django	yes
django__django-10554	django	no
django__django-10880	django	no
django__django-10914	django	yes
django__django-10973	django	yes
django__django-10999	django	no
django__django-11066	django	yes
django__django-11087	django	no
django__django-11095	django	yes
django__django-11099	django	yes
django__django-11119	django	yes
django__django-11133	django	yes

Instance	Repo	Resolved
django__django-11138	django	yes
django__django-11141	django	no
django__django-11149	django	no
django__django-11163	django	yes
django__django-11179	django	yes
django__django-11206	django	yes
django__django-11211	django	no
django__django-11239	django	no
django__django-11265	django	yes
django__django-11276	django	yes
django__django-11292	django	yes
django__django-11299	django	yes
django__django-11333	django	yes
django__django-11400	django	no
django__django-11433	django	yes
django__django-11451	django	yes
django__django-11477	django	yes
django__django-11490	django	yes
django__django-11532	django	yes
django__django-11551	django	yes
django__django-11555	django	yes
django__django-11603	django	yes
django__django-11728	django	yes
django__django-11734	django	yes
django__django-11740	django	yes
django__django-11749	django	yes
django__django-11790	django	no
django__django-11815	django	yes
django__django-11820	django	no
django__django-11848	django	no
django__django-11880	django	yes
django__django-11885	django	yes
django__django-11951	django	yes
django__django-11964	django	yes
django__django-11999	django	yes
django__django-12039	django	yes
django__django-12050	django	yes
django__django-12125	django	yes
django__django-12143	django	yes
django__django-12155	django	yes
django__django-12193	django	yes
django__django-12209	django	yes
django__django-12262	django	yes
django__django-12273	django	yes
django__django-12276	django	yes
django__django-12304	django	yes
django__django-12308	django	no
django__django-12325	django	no
django__django-12406	django	no

Instance	Repo	Resolved
django__django-12419	django	yes
django__django-12663	django	yes
django__django-12708	django	no
django__django-12713	django	yes
django__django-12741	django	yes
django__django-12754	django	yes
django__django-12774	django	no
django__django-12858	django	yes
django__django-12965	django	yes
django__django-13012	django	no
django__django-13023	django	no
django__django-13028	django	yes
django__django-13033	django	yes
django__django-13089	django	no
django__django-13109	django	yes
django__django-13112	django	yes
django__django-13121	django	yes
django__django-13128	django	yes
django__django-13158	django	yes
django__django-13195	django	yes
django__django-13212	django	no
django__django-13279	django	yes
django__django-13297	django	yes
django__django-13315	django	yes
django__django-13343	django	yes
django__django-13344	django	yes
django__django-13346	django	yes
django__django-13363	django	yes
django__django-13401	django	no
django__django-13406	django	yes
django__django-13410	django	yes
django__django-13417	django	yes
django__django-13449	django	yes
django__django-13512	django	yes
django__django-13513	django	yes
django__django-13516	django	yes
django__django-13551	django	no
django__django-13568	django	no
django__django-13569	django	yes
django__django-13590	django	no
django__django-13658	django	yes
django__django-13670	django	no
django__django-13741	django	yes
django__django-13786	django	yes
django__django-13794	django	yes
django__django-13807	django	no
django__django-13809	django	yes
django__django-13810	django	yes
django__django-13820	django	no

Instance	Repo	Resolved
django__django-13821	django	yes
django__django-13837	django	yes
django__django-13925	django	yes
django__django-13933	django	yes
django__django-13964	django	yes
django__django-14007	django	yes
django__django-14011	django	no
django__django-14017	django	yes
django__django-14034	django	yes
django__django-14053	django	yes
django__django-14089	django	yes
django__django-14122	django	yes
django__django-14140	django	yes
django__django-14155	django	yes
django__django-14170	django	no
django__django-14238	django	yes
django__django-14311	django	no
django__django-14315	django	yes
django__django-14349	django	yes
django__django-14351	django	yes
django__django-14373	django	yes
django__django-14376	django	yes
django__django-14404	django	yes
django__django-14434	django	yes
django__django-14493	django	yes
django__django-14500	django	yes
django__django-14534	django	yes
django__django-14539	django	yes
django__django-14559	django	yes
django__django-14580	django	yes
django__django-14608	django	yes
django__django-14631	django	yes
django__django-14672	django	yes
django__django-14725	django	yes
django__django-14752	django	yes
django__django-14765	django	yes
django__django-14771	django	yes
django__django-14787	django	yes
django__django-14792	django	yes
django__django-14855	django	yes
django__django-14915	django	yes
django__django-14999	django	yes
django__django-15022	django	yes
django__django-15037	django	yes
django__django-15098	django	yes
django__django-15103	django	yes
django__django-15104	django	yes
django__django-15127	django	yes
django__django-15128	django	yes

Instance	Repo	Resolved
django__django-15161	django	yes
django__django-15252	django	yes
django__django-15268	django	yes
django__django-15277	django	yes
django__django-15278	django	yes
django__django-15280	django	yes
django__django-15315	django	yes
django__django-15368	django	yes
django__django-15375	django	yes
django__django-15380	django	yes
django__django-15382	django	yes
django__django-15467	django	yes
django__django-15499	django	yes
django__django-15503	django	yes
django__django-15525	django	yes
django__django-15554	django	yes
django__django-15561	django	yes
django__django-15563	django	yes
django__django-15569	django	yes
django__django-15572	django	no
django__django-15629	django	yes
django__django-15695	django	yes
django__django-15731	django	yes
django__django-15732	django	yes
django__django-15741	django	yes
django__django-15814	django	yes
django__django-15851	django	yes
django__django-15863	django	yes
django__django-15916	django	yes
django__django-15930	django	yes
django__django-15957	django	yes
django__django-15973	django	yes
django__django-15987	django	yes
django__django-16032	django	yes
django__django-16082	django	yes
django__django-16100	django	yes
django__django-16116	django	yes
django__django-16136	django	yes
django__django-16139	django	yes
django__django-16145	django	no
django__django-16255	django	yes
django__django-16256	django	yes
django__django-16263	django	yes
django__django-16315	django	yes
django__django-16333	django	yes
django__django-16429	django	yes
django__django-16454	django	yes
django__django-16485	django	yes
django__django-16493	django	yes

Instance	Repo	Resolved
django__django-16502	django	yes
django__django-16527	django	yes
django__django-16560	django	yes
django__django-16569	django	yes
django__django-16595	django	yes
django__django-16612	django	yes
django__django-16631	django	yes
django__django-16642	django	yes
django__django-16661	django	yes
django__django-16662	django	yes
django__django-16667	django	yes
django__django-16801	django	yes
django__django-16819	django	yes
django__django-16877	django	yes
django__django-16899	django	yes
django__django-16901	django	yes
django__django-16938	django	yes
django__django-16950	django	yes
django__django-17029	django	yes
django__django-17084	django	yes
django__django-17087	django	yes
django__django-7530	django	yes
django__django-9296	django	yes
matplotlib__matplotlib-13989	matplotlib	yes
matplotlib__matplotlib-14623	matplotlib	yes
matplotlib__matplotlib-20488	matplotlib	yes
matplotlib__matplotlib-20676	matplotlib	yes
matplotlib__matplotlib-20826	matplotlib	yes
matplotlib__matplotlib-20859	matplotlib	yes
matplotlib__matplotlib-21568	matplotlib	no
matplotlib__matplotlib-22719	matplotlib	yes
matplotlib__matplotlib-22865	matplotlib	yes
matplotlib__matplotlib-22871	matplotlib	yes
matplotlib__matplotlib-23299	matplotlib	yes
matplotlib__matplotlib-23314	matplotlib	yes
matplotlib__matplotlib-23412	matplotlib	yes
matplotlib__matplotlib-23476	matplotlib	yes
matplotlib__matplotlib-24026	matplotlib	yes
matplotlib__matplotlib-24149	matplotlib	yes
matplotlib__matplotlib-24177	matplotlib	yes
matplotlib__matplotlib-24570	matplotlib	yes
matplotlib__matplotlib-24627	matplotlib	no
matplotlib__matplotlib-24637	matplotlib	yes
matplotlib__matplotlib-24870	matplotlib	yes
matplotlib__matplotlib-24970	matplotlib	yes
matplotlib__matplotlib-25122	matplotlib	yes
matplotlib__matplotlib-25287	matplotlib	yes
matplotlib__matplotlib-25311	matplotlib	yes
matplotlib__matplotlib-25332	matplotlib	yes

Instance	Repo	Resolved
matplotlib__matplotlib-25479	matplotlib	no
matplotlib__matplotlib-25775	matplotlib	yes
matplotlib__matplotlib-25960	matplotlib	no
matplotlib__matplotlib-26113	matplotlib	yes
matplotlib__matplotlib-26208	matplotlib	yes
matplotlib__matplotlib-26291	matplotlib	yes
matplotlib__matplotlib-26342	matplotlib	yes
matplotlib__matplotlib-26466	matplotlib	yes
mwaskom__seaborn-3069	mwaskom	yes
mwaskom__seaborn-3187	mwaskom	yes
pallets__flask-5014	pallets	yes
psf__requests-1142	psf	yes
psf__requests-1724	psf	yes
psf__requests-1766	psf	yes
psf__requests-1921	psf	yes
psf__requests-2317	psf	yes
psf__requests-2931	psf	no
psf__requests-5414	psf	yes
psf__requests-6028	psf	yes
pydata__xarray-2905	pydata	yes
pydata__xarray-3095	pydata	yes
pydata__xarray-3151	pydata	yes
pydata__xarray-3305	pydata	yes
pydata__xarray-3677	pydata	yes
pydata__xarray-3993	pydata	yes
pydata__xarray-4075	pydata	yes
pydata__xarray-4094	pydata	yes
pydata__xarray-4356	pydata	yes
pydata__xarray-4629	pydata	yes
pydata__xarray-4687	pydata	yes
pydata__xarray-4695	pydata	yes
pydata__xarray-4966	pydata	yes
pydata__xarray-6461	pydata	yes
pydata__xarray-6599	pydata	yes
pydata__xarray-6721	pydata	yes
pydata__xarray-6744	pydata	yes
pydata__xarray-6938	pydata	yes
pydata__xarray-6992	pydata	no
pydata__xarray-7229	pydata	no
pydata__xarray-7233	pydata	yes
pydata__xarray-7393	pydata	yes
pylint-dev__pylint-4551	pylint-dev	yes
pylint-dev__pylint-4604	pylint-dev	no
pylint-dev__pylint-4661	pylint-dev	yes
pylint-dev__pylint-4970	pylint-dev	yes
pylint-dev__pylint-6386	pylint-dev	yes
pylint-dev__pylint-6528	pylint-dev	no
pylint-dev__pylint-6903	pylint-dev	yes
pylint-dev__pylint-7080	pylint-dev	yes

Instance	Repo	Resolved
pylint-dev__pylint-7277	pylint-dev	yes
pylint-dev__pylint-8898	pylint-dev	no
pytest-dev__pytest-10051	pytest-dev	yes
pytest-dev__pytest-10081	pytest-dev	yes
pytest-dev__pytest-10356	pytest-dev	yes
pytest-dev__pytest-5262	pytest-dev	yes
pytest-dev__pytest-5631	pytest-dev	yes
pytest-dev__pytest-5787	pytest-dev	no
pytest-dev__pytest-5809	pytest-dev	yes
pytest-dev__pytest-5840	pytest-dev	yes
pytest-dev__pytest-6197	pytest-dev	yes
pytest-dev__pytest-6202	pytest-dev	yes
pytest-dev__pytest-7205	pytest-dev	yes
pytest-dev__pytest-7236	pytest-dev	yes
pytest-dev__pytest-7324	pytest-dev	yes
pytest-dev__pytest-7432	pytest-dev	yes
pytest-dev__pytest-7490	pytest-dev	yes
pytest-dev__pytest-7521	pytest-dev	yes
pytest-dev__pytest-7571	pytest-dev	yes
pytest-dev__pytest-7982	pytest-dev	yes
pytest-dev__pytest-8399	pytest-dev	yes
scikit-learn__scikit-learn-10297	scikit-learn	yes
scikit-learn__scikit-learn-10844	scikit-learn	yes
scikit-learn__scikit-learn-10908	scikit-learn	yes
scikit-learn__scikit-learn-11310	scikit-learn	yes
scikit-learn__scikit-learn-11578	scikit-learn	yes
scikit-learn__scikit-learn-12585	scikit-learn	yes
scikit-learn__scikit-learn-12682	scikit-learn	yes
scikit-learn__scikit-learn-12973	scikit-learn	yes
scikit-learn__scikit-learn-13124	scikit-learn	yes
scikit-learn__scikit-learn-13135	scikit-learn	yes
scikit-learn__scikit-learn-13142	scikit-learn	yes
scikit-learn__scikit-learn-13328	scikit-learn	yes
scikit-learn__scikit-learn-13439	scikit-learn	yes
scikit-learn__scikit-learn-13496	scikit-learn	yes
scikit-learn__scikit-learn-13779	scikit-learn	yes
scikit-learn__scikit-learn-14053	scikit-learn	yes
scikit-learn__scikit-learn-14087	scikit-learn	no
scikit-learn__scikit-learn-14141	scikit-learn	yes
scikit-learn__scikit-learn-14496	scikit-learn	yes
scikit-learn__scikit-learn-14629	scikit-learn	yes
scikit-learn__scikit-learn-14710	scikit-learn	yes
scikit-learn__scikit-learn-14894	scikit-learn	yes
scikit-learn__scikit-learn-14983	scikit-learn	no
scikit-learn__scikit-learn-15100	scikit-learn	yes
scikit-learn__scikit-learn-25102	scikit-learn	yes
scikit-learn__scikit-learn-25232	scikit-learn	yes
scikit-learn__scikit-learn-25747	scikit-learn	yes
scikit-learn__scikit-learn-25931	scikit-learn	yes

Instance	Repo	Resolved
scikit-learn__scikit-learn-25973	scikit-learn	yes
scikit-learn__scikit-learn-26194	scikit-learn	yes
scikit-learn__scikit-learn-26323	scikit-learn	yes
scikit-learn__scikit-learn-9288	scikit-learn	yes
sphinx-doc__sphinx-10323	sphinx-doc	yes
sphinx-doc__sphinx-10435	sphinx-doc	no
sphinx-doc__sphinx-10449	sphinx-doc	yes
sphinx-doc__sphinx-10466	sphinx-doc	yes
sphinx-doc__sphinx-10614	sphinx-doc	no
sphinx-doc__sphinx-10673	sphinx-doc	yes
sphinx-doc__sphinx-11445	sphinx-doc	yes
sphinx-doc__sphinx-11510	sphinx-doc	no
sphinx-doc__sphinx-7440	sphinx-doc	yes
sphinx-doc__sphinx-7454	sphinx-doc	yes
sphinx-doc__sphinx-7462	sphinx-doc	yes
sphinx-doc__sphinx-7590	sphinx-doc	yes
sphinx-doc__sphinx-7748	sphinx-doc	yes
sphinx-doc__sphinx-7757	sphinx-doc	yes
sphinx-doc__sphinx-7889	sphinx-doc	yes
sphinx-doc__sphinx-7910	sphinx-doc	yes
sphinx-doc__sphinx-7985	sphinx-doc	yes
sphinx-doc__sphinx-8035	sphinx-doc	yes
sphinx-doc__sphinx-8056	sphinx-doc	yes
sphinx-doc__sphinx-8120	sphinx-doc	yes
sphinx-doc__sphinx-8265	sphinx-doc	no
sphinx-doc__sphinx-8269	sphinx-doc	yes
sphinx-doc__sphinx-8459	sphinx-doc	yes
sphinx-doc__sphinx-8475	sphinx-doc	yes
sphinx-doc__sphinx-8548	sphinx-doc	yes
sphinx-doc__sphinx-8551	sphinx-doc	yes
sphinx-doc__sphinx-8593	sphinx-doc	yes
sphinx-doc__sphinx-8595	sphinx-doc	yes
sphinx-doc__sphinx-8621	sphinx-doc	yes
sphinx-doc__sphinx-8638	sphinx-doc	yes
sphinx-doc__sphinx-8721	sphinx-doc	yes
sphinx-doc__sphinx-9229	sphinx-doc	yes
sphinx-doc__sphinx-9230	sphinx-doc	yes
sphinx-doc__sphinx-9258	sphinx-doc	yes
sphinx-doc__sphinx-9281	sphinx-doc	yes
sphinx-doc__sphinx-9320	sphinx-doc	yes
sphinx-doc__sphinx-9367	sphinx-doc	yes
sphinx-doc__sphinx-9461	sphinx-doc	yes
sphinx-doc__sphinx-9591	sphinx-doc	yes
sphinx-doc__sphinx-9602	sphinx-doc	yes
sphinx-doc__sphinx-9658	sphinx-doc	no
sphinx-doc__sphinx-9673	sphinx-doc	yes
sphinx-doc__sphinx-9698	sphinx-doc	yes
sphinx-doc__sphinx-9711	sphinx-doc	no
sympy__sympy-11618	sympy	yes

Instance	Repo	Resolved
sympy__sympy-12096	sympy	yes
sympy__sympy-12419	sympy	yes
sympy__sympy-12481	sympy	no
sympy__sympy-12489	sympy	yes
sympy__sympy-13031	sympy	no
sympy__sympy-13091	sympy	yes
sympy__sympy-13372	sympy	no
sympy__sympy-13480	sympy	yes
sympy__sympy-13551	sympy	yes
sympy__sympy-13615	sympy	yes
sympy__sympy-13647	sympy	yes
sympy__sympy-13757	sympy	yes
sympy__sympy-13798	sympy	yes
sympy__sympy-13852	sympy	no
sympy__sympy-13877	sympy	yes
sympy__sympy-13878	sympy	yes
sympy__sympy-13974	sympy	yes
sympy__sympy-14248	sympy	no
sympy__sympy-14531	sympy	yes
sympy__sympy-14711	sympy	yes
sympy__sympy-14976	sympy	yes
sympy__sympy-15017	sympy	yes
sympy__sympy-15345	sympy	yes
sympy__sympy-15349	sympy	yes
sympy__sympy-15599	sympy	no
sympy__sympy-15809	sympy	yes
sympy__sympy-15875	sympy	yes
sympy__sympy-15976	sympy	no
sympy__sympy-16450	sympy	yes
sympy__sympy-16597	sympy	no
sympy__sympy-16766	sympy	yes
sympy__sympy-16792	sympy	yes
sympy__sympy-16886	sympy	yes
sympy__sympy-17139	sympy	yes
sympy__sympy-17318	sympy	no
sympy__sympy-17630	sympy	yes
sympy__sympy-17655	sympy	yes
sympy__sympy-18189	sympy	yes
sympy__sympy-18199	sympy	yes
sympy__sympy-18211	sympy	yes
sympy__sympy-18698	sympy	no
sympy__sympy-18763	sympy	no
sympy__sympy-19040	sympy	yes
sympy__sympy-19346	sympy	yes
sympy__sympy-19495	sympy	yes
sympy__sympy-19637	sympy	yes
sympy__sympy-19783	sympy	yes
sympy__sympy-19954	sympy	yes
sympy__sympy-20154	sympy	yes

Instance	Repo	Resolved
sympy__sympy-20428	sympy	yes
sympy__sympy-20438	sympy	no
sympy__sympy-20590	sympy	yes
sympy__sympy-20801	sympy	yes
sympy__sympy-20916	sympy	yes
sympy__sympy-21379	sympy	yes
sympy__sympy-21596	sympy	no
sympy__sympy-21612	sympy	no
sympy__sympy-21847	sympy	yes
sympy__sympy-21930	sympy	yes
sympy__sympy-22080	sympy	no
sympy__sympy-22456	sympy	yes
sympy__sympy-22714	sympy	yes
sympy__sympy-22914	sympy	yes
sympy__sympy-23262	sympy	yes
sympy__sympy-23413	sympy	no
sympy__sympy-23534	sympy	yes
sympy__sympy-23824	sympy	yes
sympy__sympy-23950	sympy	no
sympy__sympy-24066	sympy	yes
sympy__sympy-24213	sympy	yes
sympy__sympy-24443	sympy	yes
sympy__sympy-24539	sympy	yes
sympy__sympy-24562	sympy	yes
sympy__sympy-24661	sympy	yes

C Multilingual Summary

This report is accompanied by a multilingual README set in the artifact repository, so that readers of many languages can understand the system and verify the result. Each README states the headline number, the integrity constraints, and the three verification paths. The languages covered are: Bengali, Chinese, Hindi, Spanish, Arabic, French, Russian, Portuguese, German, Japanese, Korean, Indonesian, Urdu, Turkish, Persian, Vietnamese, Thai, Italian, Dutch, Polish, and English.